

The Energy Balance Project

Homeostatic Field Model and Local Actor-Motion Law

Foundation Document — Version 2.4 (Foresight vs Artifact; Protective Harvest Rules)

Status: the V2.3 conclusion is refined by the control it was missing, and three protective harvest rules are added and compared. Backed by scripts and a 24-test suite.

Central correction. V2.3 reported that long-horizon foresight destroyed a regenerating resource. That was imprecise: the horizon test there only *gated* a quantity chosen by the immediate ($H=1$) line search. When the quantity is instead chosen to **maximise** the H-tick impact ($q_{H^*} = \operatorname{argmax}_q I^H(q)$), harvesting is fully sustainable (0/32 sources lost, 100% of demand served). So the V2.3 collapse was an **artifact of gating an immediately-optimised quantity**, not evidence that foresight is inherently destructive.

Practical finding. A cheap threshold-aware burden term matches the expensive horizon-optimised rule (both sustainable, demand fully served) at $\sim 1/100$ th the cost, and causes zero regeneration-threshold crossings. And on genuinely randomised clustered layouts, foresight robustly improves routing (+22.3 points over the myopic rule, winning all 20 paired layouts).

1. The Control V2.3 Lacked

In V2.3 the regeneration-aware actor (Model D) selected the transfer amount with the immediate burden line search and then used the H-tick counterfactual only to accept or reject it. The acceptance test passes any q whose predicted impact is positive, including an over-large immediately-optimal q . The correct, narrower statement of the V2.3 result is therefore:

A long isolated-action ACCEPTANCE test applied to an immediately-optimised harvest quantity can encourage destructive harvesting.

This does not establish that foresight itself is harmful. To decide that, the quantity must itself be chosen by the horizon objective.

2. Three Protective Rules and the Decisive Control

2.1 Threshold-aware burden (regenerative reserve)

Give each regenerative source a reserve R_i above its Allee threshold and penalise dropping below it, preserving the convex piecewise-quadratic structure the line search needs:

```
R_i = A_i + delta_i
B_regen = B + sum_i chi_i [R_i - x_i]_+^2
mu_i^regen adds -2 chi_i (R_i - x_i) for x_i < R_i (a depleted source now 'pulls')
```

2.2 Hard-reserve baseline

A blunt but clear baseline: cap harvesting at the surplus above the reserve, and forbid it while the source is below its Allee threshold.

```
q_max^sustainable = max(0, x_i - R_i); harvesting forbidden while x_i < A_i
```

2.3 Horizon-optimised quantity (the control)

Rather than choose q immediately and ask the horizon yes/no, choose q to maximise the horizon objective directly:

```
q_H* = argmax_{0 <= q <= q_physical} I^H(q)
```

If this still over-harvests, the single-action counterfactual is fundamentally biased; if it is sustainable, only the accept/reject architecture was at fault.

3. Regeneration Results

Closed Allee economy (8x8, only supply is regeneration, $A=8 > L=4$), 1000 ticks, a 55% source shock at tick 500. Success requires preserving sources AND serving demand.

Rule	viable %	demand served %	dead sources	A-crossings	src stock	cum. burden
safe (H=1)	100.0	100.0	0 / 32	1	587	94,380
horizon_gate (V2.3)	42.8	73.8	32 / 32	60	0	326,941
horizon_opt (q_H*)	100.0	100.0	0 / 32	2	585	110,853
threshold_penalty	100.0	100.0	0 / 32	0	587	95,118
hard_reserve	100.0	100.0	0 / 32	1	587	94,380
penalty_horizon	100.0	100.0	0 / 32	0	585	111,930

Only horizon_gate fails. The horizon-optimised rule is sustainable, proving foresight is not inherently destructive. The two burden-based rules (threshold_penalty, penalty_horizon) achieve zero threshold crossings — they never even approach the Allee boundary — and the cheap threshold_penalty matches the ~100x more expensive horizon_opt. All sustainable rules serve 100% of demand, so none succeeds by simply refusing to harvest.

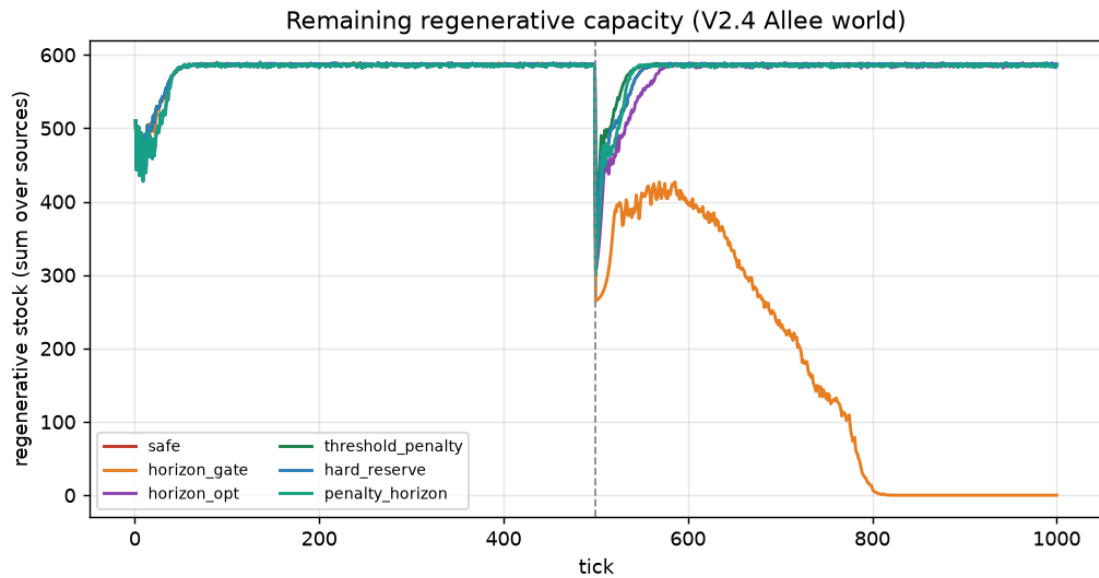


Figure 1. Remaining regenerative stock. After the shock only horizon_gate (orange) collapses; the horizon-optimised and burden-based rules all recover fully.

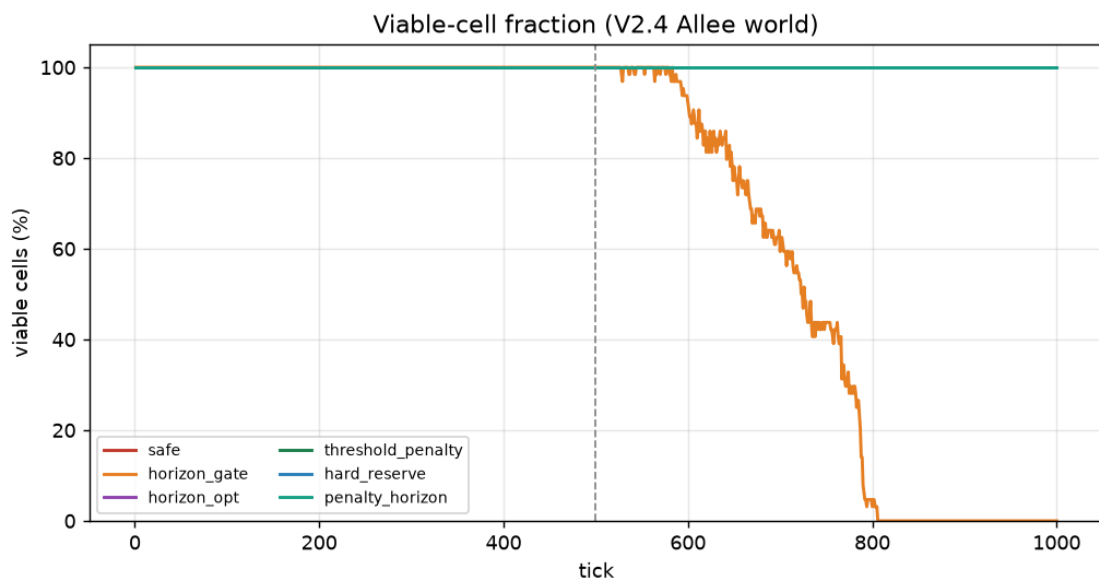


Figure 2. Viable-cell fraction. Five of six rules hold ~100%; only the V2.3 gate collapses.

4. Randomised Clustered Audit

The V2.3 clustered layout was deterministic. Here every seed draws a distinct world (random cluster count, centre, radius, density) and every model runs on the identical world for that seed (paired). Producers are external-flow sources, so the regeneration rules do not apply; we compare transport rules.

Rule	viable % (mean $\pm$ sd)	min	max	seeds
none	22.0 $\pm$ 11.1	9.4	54.7	20

gradient	35.4 ± 12.8	12.1	64.0	20
safe (H=1)	35.5 ± 13.5	14.3	63.2	20
horizon_gate (H=10)	57.9 ± 11.4	34.9	85.2	20
horizon_opt (H=10)	58.8 ± 14.5	44.0	86.3	6

The Stage-1 (V2.2.1) finding survives layout randomisation: foresight substantially improves clustered routing. In the paired comparison, horizon_gate beats the myopic safe rule by +22.3 points on average and wins all 20/20 layouts. (horizon_opt was run on 6 seeds for cost.)

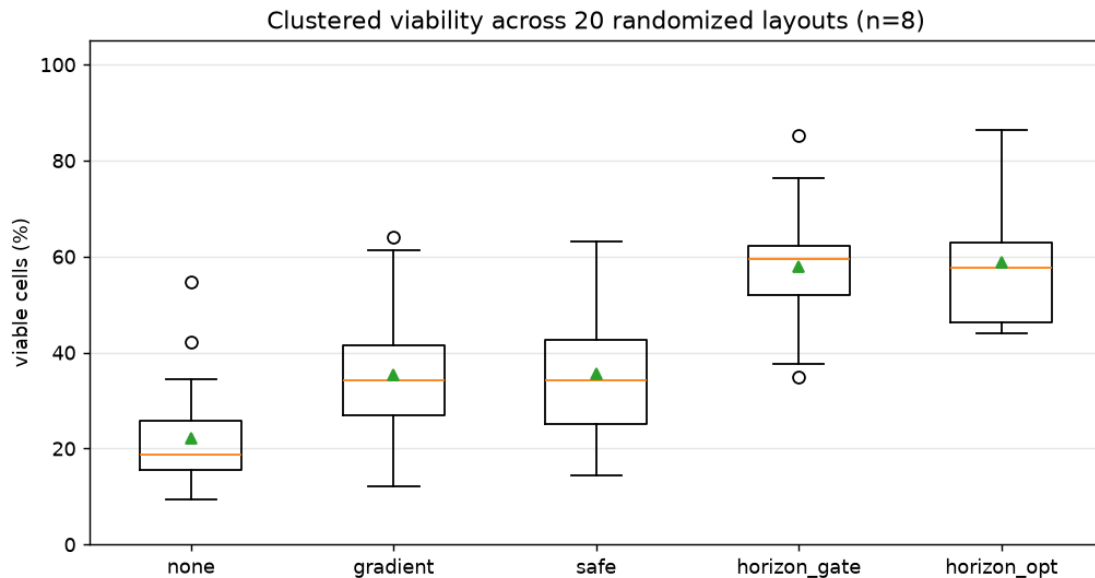


Figure 3. Distribution of clustered viability across 20 randomised paired layouts. Foresight (horizon_gate / horizon_opt) clearly shifts the distribution upward.

5. Verdict

Foresight is not the villain; the accept/reject architecture was. The precise lesson of V2.3–V2.4 is that pairing a long isolated-action acceptance test with an immediately-optimised quantity encourages over-harvest, but the same horizon objective used to CHOOSE the quantity harvests sustainably. Foresight helps for transport routing (clustered) and, correctly formulated, does no harm for resource management.

The recommended rule is the threshold-aware burden. It preserves sources and serves all demand, produces zero Allee-threshold crossings, keeps the convex structure the line search needs, and costs ~100x less than the horizon-optimised search. This is exactly the V2.0 Sec. 7.2 design note — a critical regeneration threshold belongs in the burden and regeneration dynamics — now demonstrated rather than assumed. The hard-reserve baseline is equally safe but less flexible; the horizon rules are best reserved for transport routing, where their extra cost buys real benefit.

Success was defined as preserving sources AND serving demand, so a do-nothing rule that hoards sources while starving consumers would not have counted; every sustainable rule here serves 100% of demand.

6. Limitations and Next Steps

1. The single-action counterfactual still assumes one isolated action; horizon_opt succeeds here because the objective, maximised over q , avoids pushing the source below threshold, but in adversarial or strategic settings a mean-field or repeated-action model may be needed.

2. horizon_opt used a coarse grid + one refinement; a finer optimiser might shift its numbers slightly (not its qualitative sustainability).
3. Clustered horizon_opt was only 6 seeds for cost; extend for tighter intervals.
4. With a trusted threshold-aware rule in hand, the EBU accounting overlay can finally be added and tested for gaming, hoarding, and repeated credit.

Reproducibility. Pure-stdlib core: energy_balance.py (+ tests). Engines: ebu_v22.py, ebu_v23.py, ebu_v24.py. Experiments: ecosystem.py, experiments_v22.py, audit_v231.py, exp_v23.py, exp_v24.py, exp_v24_clustered.py. Tests (24 total): test_energy_balance.py (8), test_v22.py (7), test_v23.py (4), test_v24.py (5). Figures and PDFs are regenerated by the make_paper_*.py scripts. The figure/PDF scripts require matplotlib + reportlab; the core and its physics tests need only the standard library.